UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2008 question paper

5090 BIOLOGY

5090/03 Paper 3 (Practical Test), maximum raw mark 40

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1 Table 1.1 completed realistically to 10 min ;
alternating correctly;
starting (of both) and final air temperatures recorded, with units; [3]

(a) (i) Graph marks:
1. axes labelled 'time/min' & 'temp/°C'
   [N.B. correct origin for *y*-axis]
2. orientation (*x* – horizontal), suitable size (> 6 × 6 cm) and scale
   [at least 1 cm between curves]
3. plots clear and accurate
4. both curves well done (ruled or best fit curve)
5. curves distinguished and identified [5]

(ii) still diverging;
(both) levelling out/constant/reduced gradient;
(both) decreasing;
B always above **A**; [max 2]

(b) (i) evaporation;
latent heat;
(heat) from cup/water; [max 2]

(ii) (sweat) glands (secrete sweat);
evaporation takes heat/causes cooling;
cooling skin/blood/body; [max 2]

(iii) variable sweating/receptor control AW;
ref. constant body temperature;
ref. vasodilation/ref. hair; [max 2]

(c) three from:
replicate;
stir contents (with thermometer);
lid with hole allowing it to remain in place;
2 thermometers;
qualified wetting (time, volume, etc.);
equal volumes of water;
qualified more cups;
OVP e.g. constant environment/no draught; [3]

[Total: 19]

2 (a) (i) drawing marks:
1. at least 5 cm realistic, clear and clean
2. tri-partite, (good proportions)
3. vascular bundles shown in skin [3]

(ii) drawing marks:
2–5 rows of vascular bundles
inner bundles larger
indication of vessels/staining [3]

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(iii) clearly shown where measured;
both dimensions clear, units once at least;
correct expression for calculation;
correctly expressed;
up to 2 d.p. rounded up/down no more than 0.2 [4]

(b) **(i)** two from: fruit wall/pericarp/epicarp;
tripartite/other division/placentas;
vascular tissue/veins; [2]

(ii) in Table 2.1 three from:

W1	W2
wide/thick	thin/narrow;
closer/more/rows of vascular bundles	single layer/few vascular bundles;
yellow/pale green	dark and light green etc.;
angular/rough/ridged	smooth;
OVP;	

[3]

(iii) in Table 2.2 two from:

W1	W2
no/small seeds	seeds present/large;
firm/fleshy/dry	succulent/moist/jelly, etc.;
brown seeds	green/white seeds;
OVP;	

[2]

(c) bananas of different age/same banana ageing;
apply iodine test;
colour lessens AW; [R: blue]
apply Benedict's test;
more positive result/green – yellow – red, later/when riper;
verifies the hypothesis;
ref. uniform samples/test reagents;
replicates; [max 4]

[Total: 21]

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